Curriculum vitae

Last updated: 18th of December, 2025

Full name: Virinchi Venkata Rallabhandi

Website: https://virinchirallabhandi.github.io

Nationality: Australian

Current study: PhD in Mathematical Physics at the University of Edinburgh, September 2022 - April 2026 (expected submission), supervised by James Lucietti
Tentative thesis title: *A tale of spinors and positive energy theorems*

Completed qualifications:

- Master of Advanced Study in Applied Mathematics from the University of Cambridge, i.e. Part III of the Mathematics Tripos, 2021 - 2022
- Master of Physics (theoretical physics stream) from the University of Western Australia (UWA), 2020 - 2021
- Bachelor of Science (Physics, Mathematics & statistics) from UWA, 2017 - 2019

Publications:

- V. Rallabhandi. Spinorial quasilocal mass for spacetimes with negative cosmological constant, 2025. Available at arXiv[gr-qc/2504.11971] and accepted by Advances in Theoretical and Mathematical Physics.
- V. Rallabhandi. On energy bounds in asymptotically locally AdS spacetimes, 2025. Available at arXiv[gr-qc/2508.19108] and accepted by Classical and Quantum Gravity.

Talks (past and upcoming):

- Universidad Autónoma de Madrid, "SIFTS" summer school - July 19, 2024
- Beijing Institute of Mathematical Sciences and Applications, general relativity seminar - October 17, 2025. Recording available at https://bimsa.net/bimsavideo.html?id=60887.mp4.
- University of Cambridge, mathematical physics seminar - November 4, 2025
- Australian Mathematical Society annual conference - December 9, 2025
- Edinburgh Mathematical Physics Group seminar - January 28, 2026

Academic achievements:

- Equal 9th (3rd in Australia) in the pairs division (with Alexander Rohl) of the 2017 *Simon Marais Mathematics Competition* (an Asia-Pacific wide undergraduate maths competition modelled on the Putnam competition)

- Came 3rd, 1st and 3rd respectively in the 2017, 2018 and 2019 *Blakers Mathematics Competition* (a Western Australia wide undergraduate maths competition)

- Winner of the 2021 *Muriel and Colin Ramm Prize and Medal* (for the highest scoring student in the Master of Physics at UWA)

- Joint winner (with Jesse Schelfhout) of the 2019 *John de Laeter Medal* (for the highest scoring 3rd year undergraduate physics student at a Western Australian university) and winner of the 2019 *Lance Maschmedt Physics Prize* (for highest average score across the core 3rd year physics units at UWA)

- Winner of the 2019 *Ken Freeman Prize in Astrophysics and Space Science* (for highest score in the unit, PHYS3003, at UWA)

- Winner of the 2017 *Lady James Prize in Science* (for the highest score across the 1st year physics or chemistry units at UWA)

Academic experience:

- My PhD studies focus broadly on mathematical topics in general relativity. The bulk of my work so far has been concentrated on the theme of positive energy theorems and quasilocal mass. My main achievements are the proof of a positive energy theorem for asymptotically, locally AdS spacetimes with a certain class of boundary topology, the first complete proofs of BPS inequalities in this setting and the development of a new definition of quasilocal mass for spacetimes with negative cosmological constant which satisfies a number of physically desirable properties. In the few months left of my PhD, I am exploring gravitational instantons and geometric flows of quasilocal masses.

- My Cambridge master's program was entirely composed of coursework. I sat exams in general relativity, quantum field theory, differential geometry, black holes, advanced quantum field theory and string theory. Although I didn't sit exams in them, the courses I took also included symmetries, particles & fields, algebraic topology and supersymmetry.

- My UWA master's program was evenly split between coursework and research. My final thesis was titled "*Higher symmetries of relativistic wave equations in curved spacetime*." While the physical interest for studying this problem came from the parallels between the algebra of higher symmetries and higher spin algebras, my thesis was primarily concerned with the techniques of differential geometry and differential equations used to solve the problem at hand. My thesis particularly emphasized spinor methods in computing higher symmetries.

- During the 2018/2019 summer holidays, I was one of eight students selected to participate in a ten week research internship at the International Centre for Radio Astronomy Research (ICRAR) - a collaboration between physics departments at UWA and Curtin University.

Employment:

- During my PhD, I have been employed by the University of Edinburgh as a tutor for the courses, "*proofs and problem solving*," "*several variable calculus and differential equations*," "*fundamentals of pure mathematics*," "*honours analysis*," "*honours differential equations*," "*topics in mathematical physics A* [focused on black holes]," "*introduction to mathematics at university*" and "*classical field theory*," at various times.

- Between March 2020 and June 2021, I was employed by UWA to work as tutor for the first year maths unit, "*mathematical theory and methods*."

- I was employed as a tutor for the 2021 Western Australian Mathematics Summer School (WAMSS). WAMSS is a week long residential school for promising final year secondary school students, aiming to introduce them to interesting areas of maths which they wouldn't otherwise be able to access at school.

Funding:

- School of Mathematics Studentship - for my PhD at Edinburgh
- Part III International Scholarship - for my master's at Cambridge
- Muriel and Colin Ramm Postgraduate Scholarship in Physics - for my master's at UWA

Referees:

- Professor James Lucietti (j.lucietti@ed.ac.uk)
- Dr Jan Sbierski (jan.sbierski@ed.ac.uk)
- Dr Tim Adamo (t.adamo@ed.ac.uk)

Miscellaneous:

- Member of the Australian Labor Party
- Member of the Australian Republic Movement
- Life member of the Cambridge Union Society
- Long time tuba player